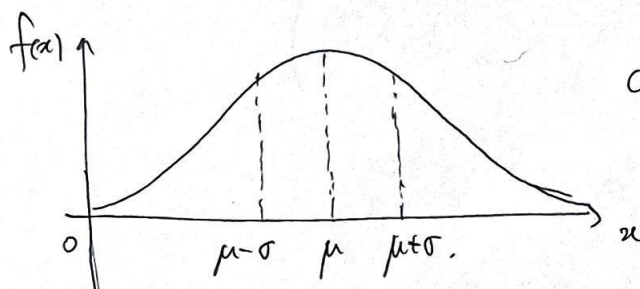


2.6.3 Normal distribution.

If a r.v. X 's pdf is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad (-\infty < x < \infty).$$

then X follows the normal distribution. Denote as $X \sim N(\mu, \sigma^2)$
 $\Rightarrow$ bell shape pdf.



convexity changes at $\mu \pm \sigma$.
 (inflexion pts)

Remark. A linear transformed random variable
 of normal dist. r.v. is also normal dist. r.v.

Let $X \sim N(\mu, \sigma^2)$, $Y = aX + b$. ($a > 0$).
 $\hookrightarrow a < 0$ (Likewise, ...).

$$\begin{aligned} \Rightarrow F_Y(y) &= P(Y \leq y) \\ &= P(aX + b \leq y) \\ &= P\left(X \leq \frac{y-b}{a}\right) \\ &= \int_{-\infty}^{\frac{y-b}{a}} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] dx \\ &= \int_{-\infty}^y \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(t-(a\mu+b))^2}{2a^2\sigma^2}\right] dt. \quad \left(\because \frac{t-b}{a} = x\right) \end{aligned}$$

$$\Rightarrow f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{\sqrt{2\pi}a\sigma} \exp\left[-\frac{(y-(a\mu+b))^2}{2a^2\sigma^2}\right] \Rightarrow \therefore Y \sim N(a\mu+b, a^2\sigma^2).$$

②

$$Z \text{ transform : } Z = \frac{X - \mu}{\sigma}$$

$$\Rightarrow f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{z^2}{2}\right], \quad (-\infty < z < \infty).$$

: Standard Normal Distribution, $N(0, 1)$.

$$\text{pdf: } \phi(z).$$

$$\text{Let } I = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$\Rightarrow I^2 = \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx \right) \left(\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy \right).$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} dx dy$$

$$\text{polar-coordinate transf. } \begin{cases} x = r \cos \theta \\ y = r \sin \theta. \end{cases}$$

$$|J| = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

$$\Rightarrow I^2 = \frac{1}{2\pi} \int_0^{2\pi} \int_0^{\infty} r \exp\left(-\frac{r^2}{2}\right) dr d\theta$$

$$= 1.$$

$$\Rightarrow I = \int_{-\infty}^{\infty} \phi(z) dz = 1.$$

Note. $X \sim N(\mu, \sigma^2) \Rightarrow E[X] = \mu, \text{Var}(X) = \sigma^2.$

$$\begin{aligned}
E[X] &= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx. \\
&= \int_{-\infty}^{\infty} (\mu + \sigma z) \phi(z) dz \quad \left(z = \frac{x-\mu}{\sigma}\right). \\
&= \mu \int_{-\infty}^{\infty} \phi(z) dz + \sigma \underbrace{\int_{-\infty}^{\infty} z \phi(z) dz}_{(\text{odd fn.})}. \\
&= \mu.
\end{aligned}$$

$$\begin{aligned}
E[X^2] &= \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx. \\
&= \int_{-\infty}^{\infty} (\mu + \sigma z)^2 \phi(z) dz. \\
&= \mu^2 \int_{-\infty}^{\infty} \phi(z) dz + 2\mu\sigma \int_{-\infty}^{\infty} z \phi(z) dz + \sigma^2 \int_{-\infty}^{\infty} z^2 \phi(z) dz. \\
&= \mu^2 + \sigma^2.
\end{aligned}$$

$$\Rightarrow \text{Var}(X) = E[X^2] - [E[X]]^2 = \sigma^2.$$

CDF of Normal distribution.

$$\Phi(x) = P(Z \leq x) = \int_{-\infty}^x \phi(t) dt.$$

Note

$$\Phi(x) = 1 - \Phi(-x).$$

④

Thm 2.20. $X \sim N(\mu, \sigma^2) \Rightarrow$ CDF of r.v. X is

$$F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

pf.)

$$F(x) = P(X \leq x)$$

$$= P\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right)$$

$$= \Phi\left(\frac{x-\mu}{\sigma}\right) \quad \left(\because \frac{X-\mu}{\sigma} \sim N(0, 1)\right) \quad \square$$

2.6.4 Bivariate Normal Distribution.Joint pdf of bivariate normal dist vector (X, Y) :

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 - 2\rho\frac{x-\mu_x}{\sigma_x}\frac{y-\mu_y}{\sigma_y} + \left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]\right].$$

$$\left\{ \begin{array}{l} -\infty < x < \infty \\ -\infty < y < \infty \\ -\infty < \mu_x, \mu_y < \infty \\ 0 < \sigma_x, \sigma_y < \infty \\ -1 < \rho < 1. \end{array} \right.$$

Denote as $(X, Y) \sim \text{BVN}(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho)$ Thm. 2.21

$$(X, Y) \sim \text{BVN}(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho).$$

$$\Rightarrow E[X] = \mu_x, E[Y] = \mu_y,$$

$$\text{Var}(X) = \sigma_x^2, \text{Var}(Y) = \sigma_y^2, \text{Cov}(X, Y) = \rho \sigma_x \sigma_y.$$

Thm. 2.22

$$(X, Y) \sim \text{BVN}(\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho)$$

$$\Rightarrow X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2).$$

pf.).

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy.$$

$$\left(v = \frac{y - \mu_Y}{\sigma_Y} \right)$$

$$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_X\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu_X}{\sigma_X} \right)^2 - \frac{1}{2(1-\rho^2)} \left(v - \rho \left(\frac{x - \mu_X}{\sigma_X} \right) \right)^2 \right] dv.$$

$$\left(w = \frac{v - \rho(x - \mu_X)/\sigma_X}{\sqrt{1-\rho^2}} \right)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp \left[-\frac{1}{2} \left(\frac{x - \mu_X}{\sigma_X} \right)^2 \right] \quad \square$$

Note. $(X, Y) \sim \text{BVN}(\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho)$

$$\Rightarrow \text{Cov}(X, Y) = 0 \rightarrow X \perp Y.$$

$\therefore \text{jpdf} = \text{marginal pdf} \times \text{marginal pdf}.$

(For Gaussian R.V.).

⑥.

Given $X=x$, the conditional pdf of Y :

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

$$= \frac{1}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2\sigma_Y^2(1-\rho^2)} \left(y - \mu_Y - \frac{\rho\sigma_Y}{\sigma_X}(x - \mu_X) \right)^2 \right]$$

$$E[Y|x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)$$

$$\text{Var}(Y|x) = \sigma_Y^2(1-\rho^2).$$

$$\Rightarrow Y|X=x \sim N \left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \sigma_Y^2(1-\rho^2) \right)$$

If $\rho=0 \Rightarrow Y|X=x$ dist. $\Leftrightarrow$ marginal dist. of Y .

If $|\rho| \uparrow \Rightarrow \sigma_Y^2(1-\rho^2) \rightarrow 0$, "degenerate distribution".